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T-cell receptor insights: Determinants of Major Histocompatibility Complex class I versus class II recognition

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Abstract

In this study, we analyzed large-scale T-cell receptor (TCR) sequence data to determine whether TCRs preferentially bind to major histocompatibility complex (MHC) class I (CD8+) or class II (CD4+) epitopes. Using the International ImMunoGeneTics information system numbering scheme, we identified specific positions with distinct amino acid enrichment for each MHC class and developed machine learning models for classification. While our frequency-based approach effectively differentiated MHC-I from MHC-II TCRs in cross-validation, performance declined when only beta chain data were used from real-world peripheral blood mononuclear cell samples. However, incorporating the TCR alpha chain significantly improved accuracy, emphasizing its importance for MHC recognition. Overall, we found that V-region loops can signal MHC class bias, aiding in immunotherapy design and TCR repertoire analysis, while highlighting the need for larger, more diverse datasets for reliable predictions.

KEYWORDS

computational biology, immunoinformatics, immunology, machine learning, T-cell receptor

1 | INTRODUCTION

T cells contribute to the adaptive immune response by recognizing various antigens (Bonilla & Oettgen, 2010; Bradley & Thomas, 2019). Two important components facilitate this response, namely the major histocompatibility complex (MHC) molecules, which present peptide fragments on the surface of nucleated cells (Neefjes et al., 2011; Wieczorek et al., 2017), and the T-cell receptors (TCR), which recognize non-self MHC-presented peptide and thereby trigger immune responses.

In humans, MHC molecules are classified into two primary classes: Human Leukocyte Antigen (HLA) class I molecules, which present endogenously derived peptides to cytotoxic T cells (CD8+ T cells) (Calis et al., 2013; Yewdell, 2022), and HLA class II molecules, which present exogenously derived peptides to helper T cells (CD4+ T cells) (Couture et al., 2019; Sant, 1994). The intricate interactions between the T cells and the peptide–MHC (pMHC) complexes ensure that the immune system can effectively combat a wide range of pathogens while maintaining tolerance to self (Messaoudi et al., 2002; Serra & Santamaria, 2019).

TCRs consist of two main chains, typically an alpha (α) and a beta (β) chain, though other combinations such as gamma (γ) and delta (δ) chains also exist (Attaf et al., 2015; Pellicci et al., 2014). Each chain contributes to the formation of the TCR's antigen-binding site. The antigen recognition capability of TCRs is primarily determined by specific regions known as complementarity-determining regions (CDRs) (Corcoran & Karlsson

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Hedestam, 2024; Gabrielli et al., 2009; Manning et al., 1999). These regions allow the T cell to specifically recognize and bind to antigens presented by MHC molecules. The CDRs are generated through the recombination of the Variable (V), Diversity (D), and Joining (J) gene segments, a genetic process that occurs during T cell development and creates a vast array of TCR diversity (Springer et al., 2021). Specifically, the CDR1 and CDR2 regions are derived from the V gene segment, while the CDR3 region arises from the junctional diversities introduced during the joining of the VDJ gene segments (Rosati et al., 2017). The CDR regions on a TCR can be seen in Figure 1.

Aside from the three main CDRs, CDR2.5 is an additional loop on the TCR. This CDR is also derived from the Variable gene, and research has found it to play a role in TCR–pMHC binding (Ehrlich et al., 2021). This loop is situated between the canonical CDR2 and CDR3 regions. In essence, CDR1, CDR2, and CDR2.5 allow for the specificity and fidelity of TCR–MHC interactions (Fernandez-Quintero et al., 2020; Wong et al., 2019), which is essential for the functioning of the adaptive immune system.

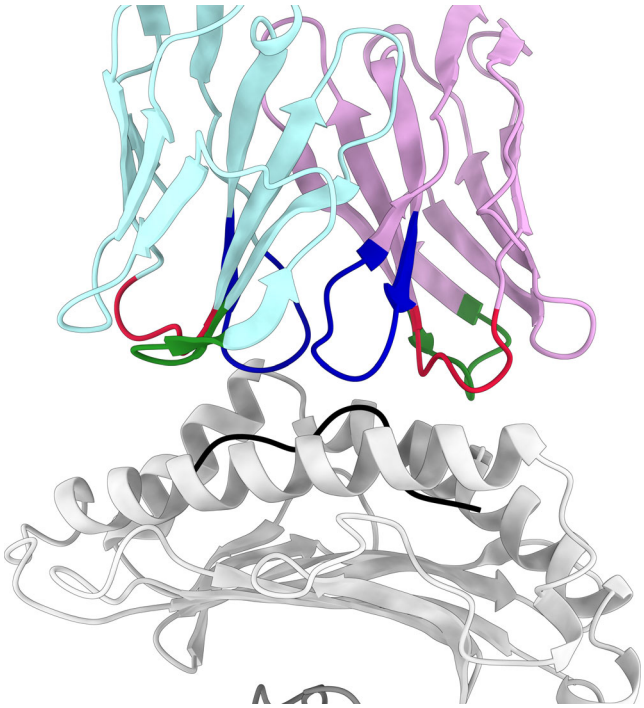

FIGURE 1 Structural representation of a T-cell receptor (TCR) highlighting the complementarity-determining regions (CDRs). The panel illustrates the TCR chains alpha (light purple) and beta (light blue) with the CDR1, CDR2, and CDR3 regions colored differently: CDR1 in crimson, CDR2 in forest green, and CDR3 in medium blue. These regions are critical for antigen (black) recognition and for binding to the MHC molecule, depicted on the bottom and colored in gray. ChimeraX (Meng et al., 2023) was used for rendering this figure, and the structure was downloaded from the protein data bank (PDB) database (Burley et al., 2023) (PDB ID: 7NME).

With the advent of large-scale immunological sequence data, repositories such as the Immune Epitope Database (IEDB) (Vita et al., 2019), VDJdb (Shugay et al., 2018), The Structural T-Cell Receptor Database (STCRDab) (Leem et al., 2018), and the adaptive immune receptor repertoire (AIRR) community (Christley et al., 2020) have significantly increased the amount of receptor data available. As a result, researchers now have access to a vast volume of information on TCR–peptide–MHC interactions.

In recent years, several software tools have been developed to examine and cluster TCR sequences, such as TCRDist (Dash et al., 2017), TCRMatch (Chronister et al., 2021), and GLIPH 2.0 (Huang et al., 2020). Additionally, tools like NetTCR (Montemurro et al., 2021), immune watch Detect (Nielsen et al., 2024), QImmuno (Mayer & Callan Jr., 2023), MixTCRpred (Croce et al., 2024), and TULIPv2 (Meynard-Piganeau et al., 2024) have been developed to predict the epitopes identified by TCR sequences. There is a competition to benchmark all the TCR predictions tools (Nielsen et al., 2024). While these tools work for epitope prediction, attributing MHC class based on TCR sequences is still a challenge. To address this issue, we developed a predictive model based on the CDR1 and CDR2 sequences of the TCR's α and β chains in order to predict MHC class from TCR sequences.

2 | RESULTS

2.1 | Differentiation in amino acid preference between MHC-I and MHC-II

We aimed to determine whether specific sequence positions in TCRs β chain exhibit differential enrichment of amino acids depending on their binding to either MHC class I or MHC class II molecules. To explore this, we analyzed the entire dataset of TCRs available in the IEDB and VDJ databases, focusing on V genes' CDRs.

To provide a standardized analysis of CDR's regions, we utilized the International ImMunoGeneTics information system (IMGT) numbering scheme (Kaas et al., 2008) to align the CDR1, CDR2, and CDR2.5 sequences to a common reference frame (for details see Section 4). This was done in order to allow comparison of amino acid composition at different positions.

As detailed in Figure 2, our analysis reveals distinct preferences for specific amino acids in various CDR regions when interacting with MHC-I and MHC-II molecules. Across different positions, we observe statistically significant differences in amino acid usage between TCRs interacting with MHC-I and MHC-II.

With its smaller peptide-binding groove flanked by α helices, MHC-I typically accommodates shorter peptides. Data show that smaller or more rigid amino acids,

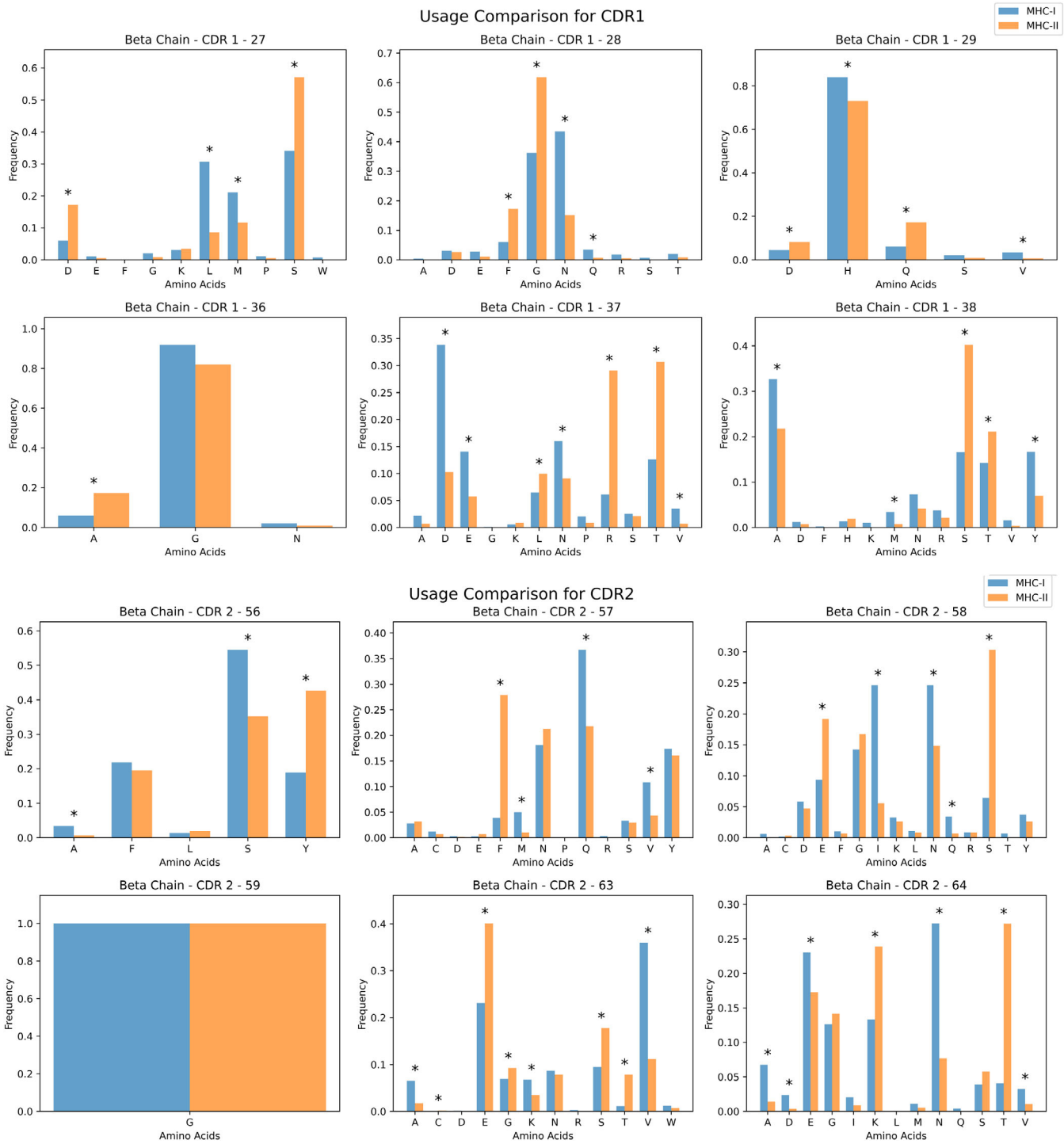

FIGURE 2 Comparative amino acid usage across complementarity-determining region (CDR) regions in T-cell receptors (TCRs) interacting with major histocompatibility complex-I (MHC-I) and MHC-II. The bar plots show the amino acid usage at each position of the CDR1, CDR2, and CDR2.5. Based on their interaction with the MHC classes, the TCRs are divided into two groups: CD8+ (blue columns) and CD4+ (orange columns). Positions where the amino acid usage significantly differs ($p < 0.05$) between CD8+ and CD4+ are marked with asterisks (see Section 4). The data reveal different patterns of amino acid selection, indicating TCR sequence variations based on the MHC type they interact with. TCRs numeric positions lacking data are omitted.

such as proline or serine, appear more frequently at positions like 37 and 56 within the CDR1 and CDR2 regions of TCRs when interacting with MHC-I. This suggests that these amino acids may play a role in the specific recognition of peptides presented by MHC-I molecules.

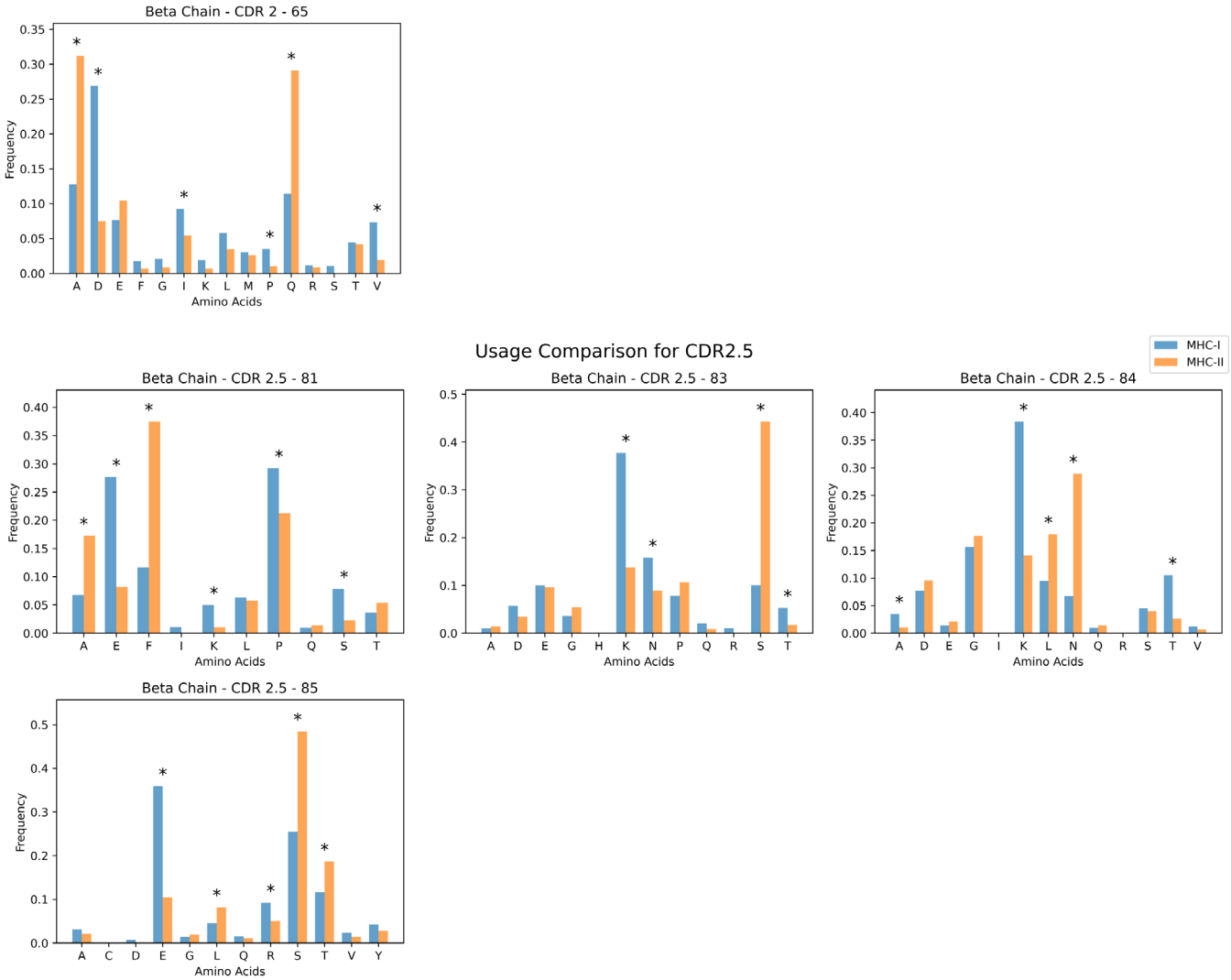

FIGURE 2 (Continued)

MHC-II molecules present longer peptides in an open and flexible groove, allowing for greater variability in TCR interactions. Positions like 28 and 57 in the CDR1 and CDR2 regions show a higher frequency of phenylalanine in interactions with MHC-II. However, while these patterns are often visible, there are exceptions where the amino acid usage in certain CDR regions does not follow these general trends. This suggests that additional factors beyond amino acid size and rigidity influence TCR and MHC interactions.

We also generated Shannon entropy plots (Figure S1), which compare MHC class I versus class II at each IMGT-defined residue position in both the α- and β-chains. These entropy plots reflect amino acid diversity per site, allowing us to see whether any single position shows markedly reduced variability (which could signal a strong contribution to MHC preference).

In general, the Shannon entropy values for each residue were not drastically different between MHC I and MHC II. Although some positions show slightly lower entropy in one class or the other, no single site consistently exhibited a major drop in variability. This finding supports the idea that MHC class usage is influenced by multiple positions rather than any single "hotspot" residue.

Our findings indicate that the difference in amino acid usage between CD4+ and CD8+ TCRs can provide good insight into TCR/MHC binding preference mechanisms; the frequency information can be utilized as input for predictive models.

2.2 | Development of a predictive model based on amino acid frequencies

We aimed to develop a prediction model to classify TCR sequences based on their MHC binding preference by retrieving the differences in amino acid usage between TCRs interacting with MHC-I and MHC-II. To

achieve this, we employed machine learning classifiers that utilize the amino acid frequencies within the CDR1 and CDR2 regions as input features. We are not including the CDR3 region because this paper focuses only on the Variable gene.

We calculated the relative frequency of each amino acid at each position within the CDR1 and CDR2 regions of TCRs. Additionally, we computed the ratios of amino acid relative frequency between TCRs interacting with MHC-I and those interacting with MHC-II at each position. These ratios highlight the difference in specific amino acid usage in one class versus the other. Lastly, we calculated the natural logarithm of the ratios.

To create a feature matrix for the set of sequences, each amino acid in the sequence was represented by its ratio between relative amino acid frequencies of the two classes. These ratios were used to create a feature matrix representing the input for our prediction models. The models utilized included linear support vector machines (linear SVC), logistic regression, and random forest classifiers. Each model was trained to distinguish CD4+ and CD8+ TCRs based on the amino acid composition at the specified CDR positions.

We implemented cross-validation using the repeated stratified *k*-fold method to split our training and testing data. Specifically, we conducted a five-fold stratified cross-validation five times, ensuring that each fold maintained the same class distribution as the entire dataset. This approach helps prevent overfitting and provides a more robust assessment of model performance. The receiver-operating characteristic (ROC) curve was generated by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings across all folds. For additional details on the cross-validation procedure, please refer to Section 4.

Our initial analysis involved comparing the predictive performance of models trained on sequences extracted from two different types of β-Variable gene assignments within the IEDB: "curated" and "calculated." Curated annotations are those reported by the original study authors. Since these TCRs come from a diverse range of laboratories and publication standards, there is considerable variability in how V genes are assigned and documented. The curated dataset has a total of 147,336 sequences. In contrast, calculated annotations rely on the automated pipeline from the IEDB. These annotations are only available for TCRs that meet specific sequence length criteria, making it a smaller set, having 12,076 sequences. In our study, we conducted a head-to-head comparison between curated and calculated TCR β genes in the context of better prediction performance, as shown in Figure 3. We also included the calculations of additional metrics, such as F1-score, precision, and recall, in Figure S2.

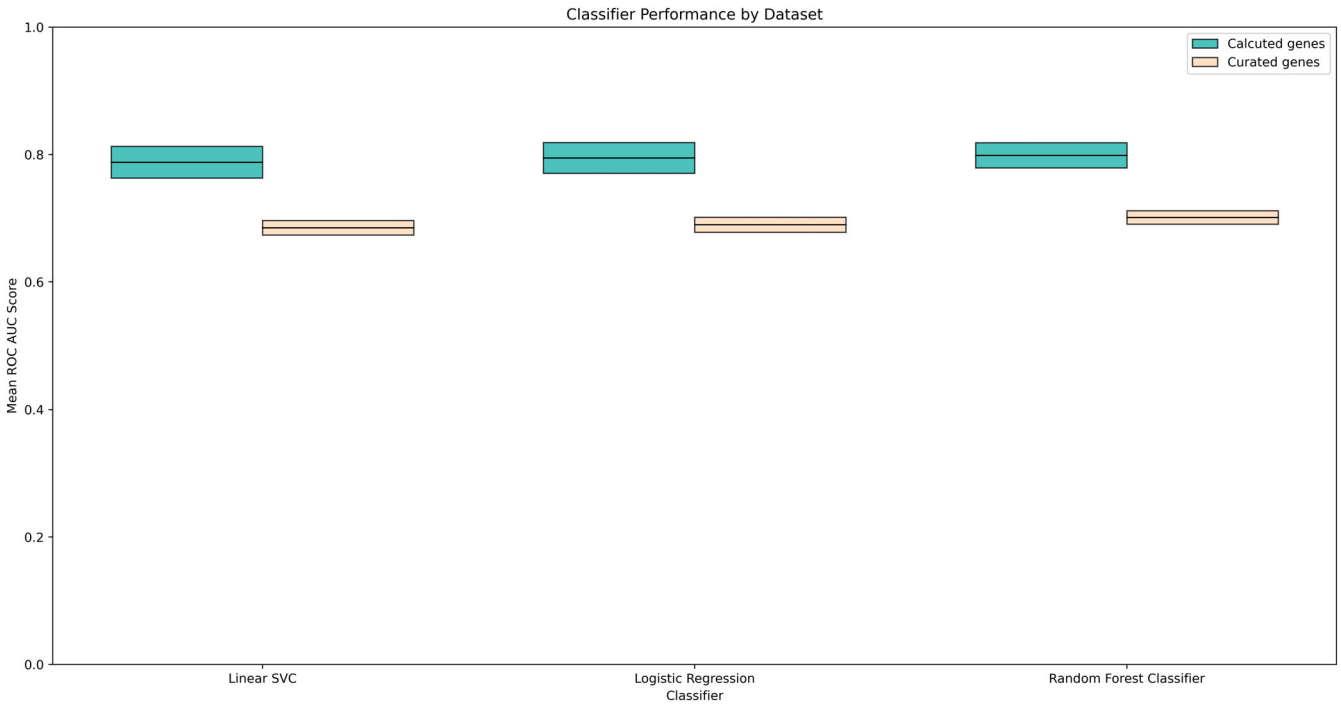

FIGURE 3 Comparison of predictive performance between curated and calculated genes from the Immune Epitope Database using three machine learning methodologies: linear support vector machines (linear SVC), logistic regression, and random forest. The floating bar plots represent each dataset and methodology's mean and standard deviation between different cross-validation folds. Notably, calculated genes showed superior performance across all models. AUC, area under the curve; ROC, receiver-operating characteristic.

The calculated genes performed better despite the manual validation and expert review associated with curated genes. This suggests that calculated genes gain performance by following the same pattern for classification, something that does not happen in curated data. Due to this performance, Figure 2 and the following results arise from an analysis that used only the calculated dataset.

We conducted a direct comparative analysis to compare the prediction between using CDR1 and CDR2 amino acid sequences compared to using only V-gene labels. We built two types of classifiers: one using solely V-gene labels as input features (e.g., "TRBV4-1" and "TRBV12-3"), and another employing the amino acid sequences of the CDR1 and CDR2 regions. As shown in Figure S3, our results indicate that while both methods achieved similar area under the receiver operating characteristic (AUROC) scores for common V genes, the sequence-based classifiers performed equally well or slightly better overall.

A significant limitation of the V-gene-only approach is that it depends exclusively on the V genes included in the training set. Due to this issue, our final predictive model incorporates the amino acid frequencies of CDR1 and CDR2.

Additionally, we performed a Leave-One-V-gene-Out (LOVO) cross-validation analysis. In this analysis, all sequences corresponding to a particular V gene were entirely excluded from training and used exclusively for testing. As shown in Figure S4, this approach strongly affected the predictive power for logistic regression; however, it did not affect the random forest. These results indicate a transferable signal embedded in the CDR1/2 sequences.

Lastly, we did a sequence similarity analysis using the basic local alignment search tool (BLAST) to determine if the direct comparison of CDR1 and CDR2 sequences could effectively classify them based on MHC binding preference. We constructed a BLAST database from the CD8+ training sequences and queried the test sequences against this database, recording the *e*-values of the top-scoring hits. The ROC curve generated from the BLAST-based predictions demonstrated poor performance, with area under the curve (AUC) values significantly lower than those obtained from our amino acid frequency-based models, as shown in Figure S5. This result suggests that only sequence similarity is insufficient for distinguishing between TCRs interacting with MHC-I and MHC-II molecules due to its lack of high variability, which only occurs in the CDR3 region. The poor performance of the BLAST approach shows the necessity of utilizing amino acid composition and frequency differences in the CDR regions to achieve better predictive accuracy.

2.3 | Incorporating structural contact information and position weighting

We compiled a structural dataset by querying the protein data bank (PDB) for all available human α-β TCR–pMHC complexes involving either MHC class I or II. After filtering, we analyzed 41 structures: 29 MHC-I and 12 MHC-II. The distribution of contacts among the TCR loops showed β-CDR2 having the highest frequency for MHC-I (36%), followed by α-CDR1 (29%) and α-CDR2 (22%), while MHC-II contacts were more evenly distributed across loops (β-CDR2: 25%, α-CDR1: 29%, α-CDR2: 22%, β-CDR1: 17%, and CDR2.5 loops ∼6%). A chi-square test comparing these distributions yielded a *p*-value of 0.0503, suggesting a borderline significant difference in loop preference between MHC-I and MHC-II interactions. We also analyzed the differences in TCR amino acid usage for CDR1 and CDR2 when comparing MHC-I and MHC-II structures and their positional contacts. However, none of these comparisons showed a statistically significant difference. The complete dataset detailing all calculated contacts is included in Table S1. The positions involved in interactions of the analyzed structures were weighted, and we incorporated this into our machine learning model.

We hypothesized that assigning different weights to these positions could enhance the model's ability to differentiate between MHC class I and II. This approach is conceptually similar to cost-sensitive learning methods used in imbalanced datasets (Araf et al., 2024), where external weights are applied to address data imbalance, potentially easing the learning and boosting the predictive performance. However, in our case, instead of adjusting weights for entire classes, we implemented feature-specific weighting.

We assigned these positions weights in our feature matrix. The weights were calculated by multiplying the natural logarithm of the amino acid frequency ratios by a factor corresponding to the contact frequency.

A particular area of interest was the loop between CDR2 and CDR3, CDR2.5. The structural analyses and literature reviews indicated this loop mainly engages with MHC molecules, as shown in Figure S6. We incorporated the frequency data of amino acid usage from CDR2.5 into our analysis.

After evaluating the models with and without the weighted positions and the inclusion of CDR2.5, as shown in Figure 4, we observed minimal differences in predictive performance.

In a further exploration of the role of individual positions and their potential impact on the TCR/MHC prediction model, we analyzed the coefficients and feature importance from various models. For instance, in the linear SVC and logistic regression models, positions 36 and 56 strongly influenced MHC-I predictions, while

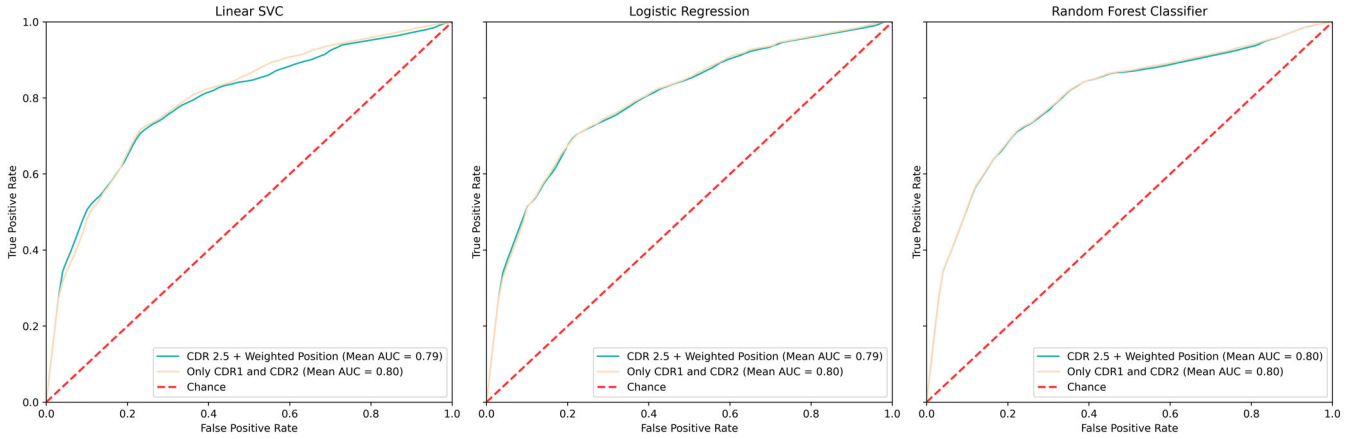

FIGURE 4 Comparison of prediction methods after including weighted and CDR2.5 positions. The panels (linear support vector machines [linear SVC), logistic regression, and random forest classifier) present graphs with two curves each. The green curve represents the scores for methods incorporating CDR2.5 and weighted positions, while the peach curve represents those without these positions. Both approaches demonstrate similar predictive capabilities. AUC, area under the curve; CDR, complementarity-determining regions.

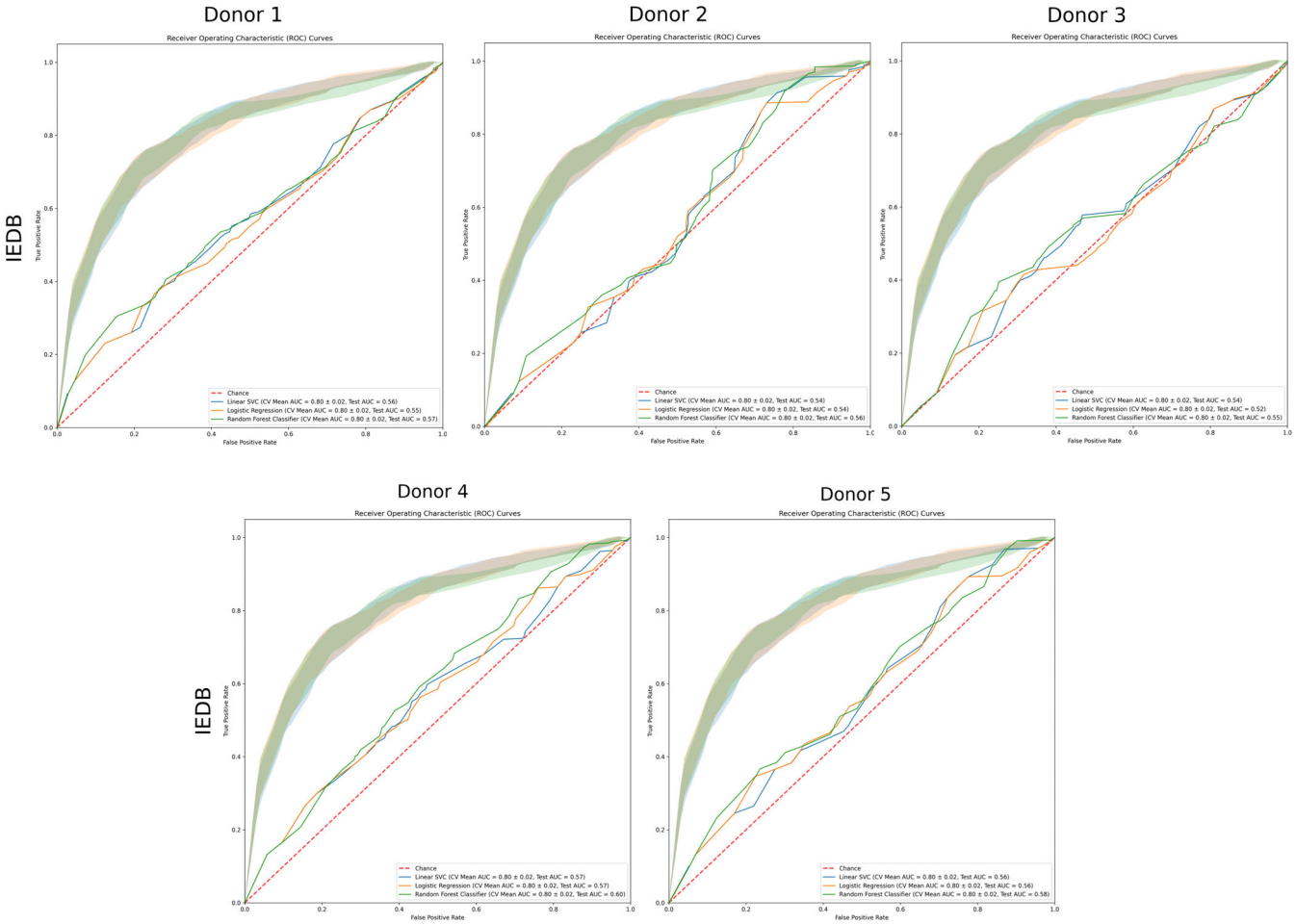

FIGURE 5 Receiver-operating characteristic (ROC) curves for the prediction models trained on the Immune Epitope Database (IEDB) dataset and tested on the peripheral blood mononuclear cells dataset from five different donors. The shaded areas represent the cross-validation result with the standard deviation of the training data, while the solid lines depict the ROC curves based on the testing data. The models show a notable decline in predictive performance when applied to the donor datasets, which is attributed to discrepancies between the training and testing datasets, including differences in CD4+/CD8+ T-cell ratios and amino acid frequency distributions. Each plot corresponds to a separate donor, illustrating the variability in prediction performance across individual donors. AUC, area under the curve; linear SVC, linear support vector machines.

positions 29, 57, and 63 showed more relevance for the MHC-II predictions. In contrast, the Random Forest Classifier emphasized new features, like positions 37 and 64, and repeated position 57 as crucial for prediction. None of the highlighted positions are from CDR2.5. These results are detailed in Table S2.

These results show that while structural contact information provides valuable biological insights, its incorporation into the model through weighting did not significantly improve accuracy.

2.4 | Integrating peripheral blood mononuclear cells data and the implications of model selection

To validate our prediction method, we next used data from a tuberculosis study developed by our research group using peripheral blood mononuclear cells (PBMCs) from five different donors. We segregated T-cells based on the presence of CD8+ and CD4+ markers, offering a natural, unexplored dataset to test the robustness of our prediction models.

We tested each donor separately, using the IEDB dataset to train the model and the donor dataset to test the prediction. An evident decline in prediction performance was observed when we applied our trained models to this unseen dataset, as seen in Figure 5.

We looked back at our data to find a reason for this decline. The first major difference is the ratio between CD4+ and CD8+ TCRs. IEDB data comprises 94% of CD8+ and 6% of CD4+. The test data, in general, are composed of 98% of CD4+ and 2% of CD8+. Our aim for using the amino acid frequency is to avoid any problem with the discrepancy between CD4+ and CD8+ since the frequency should remain at the same level.

To investigate if the frequency kept the same proportion, we checked the amino acid usage on each donor and compared it with our training data. We compared CD4+ and CD8+ training data vs. CD4+ and CD8+ for each donor in the test data, and the results are included in Figure S7. We observe several positions where the frequency of specific amino acids in the donor sequences diverges from those in the training data. For example, positions such as 37, 57, and 58 display high disparities in amino acid usage in TCRs training data, compared to the frequencies across different donors. Such discrepancies affect the model and are the likely main reasons for poor performance.

The next step was to use PBMC data from one donor to train the model and test it using PBMC data from another donor. Figure S8 shows a panel with all the results of this analysis. The performance was poor, with the ROC-AUC of the test dataset being around 0.60. The performance was consistent with when the model was trained with the IEDB dataset and tested across different donors.

2.5 | Including α information

The α chain of the TCR is critically relevant to its interaction with MHC molecules due to its direct involvement in recognizing and binding peptide–MHC complexes (La Gruta et al., 2018). To improve our method, we decided to include paired α-β data. We pulled human α-β paired data from IEDB, which contains calculated V-gene information from α and β. After filtering the results, we had 668 human TCR paired data. We retrieved paired data from the VDJdb, obtaining 27,551 MHC-I and 771 MHC-II human α-β TCRs from their database.

We applied the same approach to the α-β paired dataset. The frequency of every amino acid was calculated for every CDR1 and CDR2 position in α and β. We trained the models using VDJ data and tested them using the IEDB dataset. In Figure 6, we can see the results of this analysis. We also used the same approach but only included the β chain, as seen in Figure S9. It performed well with the IEDB testing data, outperforming the predictions that use only the β chain. These results match what exists in the literature, showing that the α chain is a key element for the recognition and interaction with MHC. We also included the calculations of additional metrics, such as F1-score, precision, and recall in Figure S2.

Random forest outperformed the other two methods. We analyzed which position was more critical for the prediction, as seen in Table S3. The most relevant positions are 28, 29, and 38 on the α chain. These indicate that this region is a hotspot for the interaction between TCR and MHC, implying the importance of the α chain's inclusion in the analysis.

We did the same approach again to check if direct sequences from α-β CDR1 and CDR2 could effectively classify TCRs based on MHC binding preferences. The results indicated that the BLAST method performed poorly on the α-β paired data too, with AUC values significantly lower than those our amino acid frequency-based models achieved, as shown on Figure S10. This suggests that mere sequence similarity is insufficient for distinguishing TCRs interacting with MHC molecules, even considering both the α and β chains.

3 | DISCUSSION/CONCLUSION

Cracking the complex interactions between TCRs and MHC molecules, specifically distinguishing TCR preferences for MHC-I versus MHC-II, represents a tricky challenge in immunology. This study's comprehensive analysis gives insights into the interaction patterns of TCR with MHC-I and MHC-II. Utilizing the CDR1, CDR2, and CDR2.5 sequences, we found that frequency-based evaluations were a good approach for

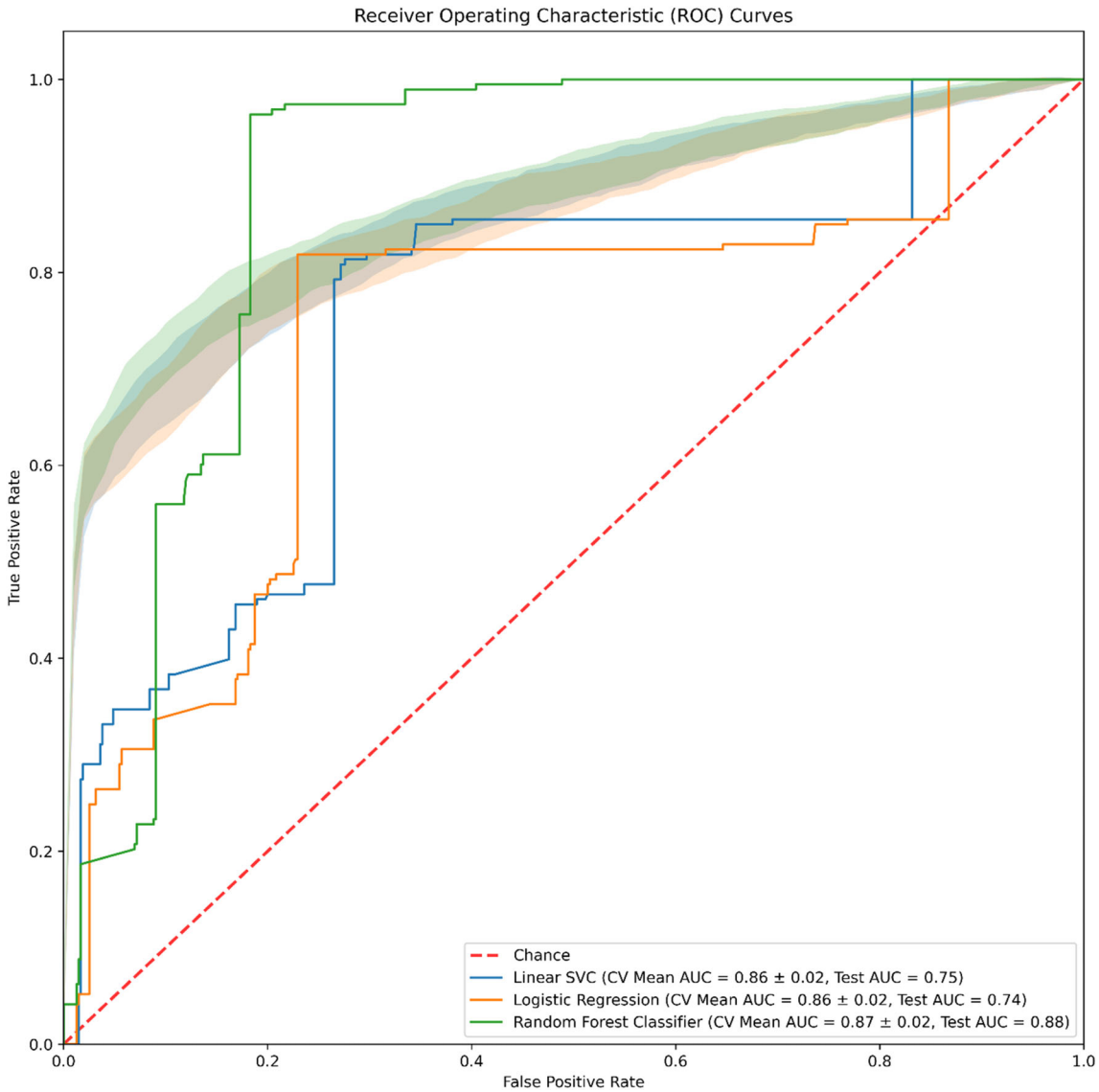

FIGURE 6 Receiver-operating characteristic (ROC) curves comparing the performance of three predictive models (linear support vector machines [linear SVC], logistic regression, and random forest classifier) trained on paired alpha (α)-beta (β) T-cell receptor (TCR) data from the VDJdb and tested on the Immune Epitope Database dataset. The shaded areas represent the cross-validation result with the standard deviation of the training data, while the solid lines depict the ROC curves based on the testing data. The random forest classifier achieved the highest performance. These results demonstrate the enhanced predictive power of models that include both the α and β chains of the TCR, corroborating the relevance of the α chain in major histocompatibility complex recognition. AUC, area under the curve.

identifying amino acid preferences and building a prediction model based on that information.

After filtering, we analyzed 41 unique human TCR–pMHC crystal structures with a resolution of a maximum of 3.0 Å, treating each complex as a single, static snapshot. Within this limited dataset, adding simple contact-count features and the full CDR 2.5 sequence did not significantly change the classifier's AUROC.

These findings do not suggest that three-dimensional information lacks predictive value; rather, they underscore the limitations of having a small and static structural sample. A deeper analysis will require molecular dynamics (MD) simulations that capture the dynamic movements of CDR 1 and CDR 2 and their transient interactions with the MHC helices. In addition to running these simulations, analyzing increased distances at which atoms are considered to be in contact, which was set in this paper at 4 Å, might be an interesting undertaking.

Our results should be interpreted as evidence that, within the structural snapshots examined in this study, additional three-dimensional features did not enhance performance compared to sequence-based models.

We encountered considerable difficulty in defining the IMGT numbered positions for the HV4a loop. We found only one resource (University of California Santa Barbara, 2012) that provided specific information about

this region, referred to as 65–72 positions in the TCRa. Using those positions, our analysis did not reveal consistent HV4–MHC contacts; so it was not included among our predictive features.

An important aspect of this work was integrating unseen external data derived from PBMCs into the analysis. The fact that performance metrics exhibited a significant decline emphasizes the challenges associated with real-world data. This highlights how hard it is to create a model generalization in personalized immunology, where donor-specific characteristics can significantly influence prediction outcomes.

Incorporating the α chain data into the analysis led to a notable enhancement in performance, clearly demonstrating the α chain's critical role in the interaction between TCRs and MHC molecules. Specifically, with paired α-β sequences, we achieved cross-validation AUCs of approximately 0.86–0.87 and external test AUCs of up to 0.88, values substantially above the random-guess baseline of 0.5. While these AUCs still fall short of ideal for clinical applications, they show that TCR–MHC class prediction is feasible and not merely random. We believe further improvements will emerge as training datasets grow and computational approaches advance.

In summary, this study introduces a method for predicting the MHC class (I or II) of TCRs based solely on their CDR1 and CDR2 sequences. We developed predictive models that demonstrated strong performance by identifying distinct amino acid preferences in TCRs that interact with MHC-I and MHC-II molecules from data. Our approach highlights the critical role of incorporating α-chain information alongside the TCR β chain, emphasizing its importance in TCR–MHC interactions. This contribution advances our understanding of T-cell recognition and offers a foundation for future research to develop targeted immunotherapies and vaccines.

4 | MATERIALS AND METHODS

4.1 | Gathering data

Within the IEDB, genes are categorized as either "calculated" or "curated." The calculated genes are derived through an automated process that uses sequence data and algorithmic predictions to determine gene classifications. Curated genes undergo meticulous review and validation by experts based on individual research studies.

From the IEDB, we specifically isolated entries of TCRs' Variable β (Vβ) gene interactions with MHC class I and II. Initially, we had 147,336 entries from human TCRs with curated genes, out of which 12,076 had associated calculated gene information.

The PBMC datasets used to validate our prediction method are from an in-house Tuberculosis study conducted at our laboratory. The T-cells were isolated from the collected samples, sequenced using a MiSeq v3 600 Cycle Full Flowcell, and grouped in CD4+ and CD8+ T-cell subsets. Around 70,000 T-cell sequence data were retrieved from each of the five donors; since it was a bulk sequencing method, we only have the β chain available.

The paired α-β TCR data utilized in this study were obtained from two sources. We retrieved 57,347 sequences from the VDJ database for the training dataset. We compiled 668 sequence data from the IEDB for the testing dataset. We only selected TCR data derived exclusively from human samples in both datasets.

4.2 | Data preprocessing

Our study prioritized the extraction of the amino acid sequence based on the VA and VB gene. To ensure standardized position references for CDR1, CDR2, and CDR2.5 across all sequences, we adopted the amino acid numbering system set by the IMGT. We used ANARCI (Dunbar & Deane, 2016) to convert TCR sequences into the format of the IMGT numbering scheme.

With its assistance, the sequence positions 27–38 were mapped and assigned to CDR1. Similarly, CDR2 was defined from positions 56–65, while the positions 81–85 corresponded to CDR2.5.

The structural set was selected based on specific requirements: (i) high-quality experimental coordinates, (ii) complete α and β TCR chains, (iii) complete peptide and MHC helices, and (iv) no redundancy in the TCR sequences. To achieve this, we followed these steps:

1. Obtained a list of TCR PDB entries from the IEDB—440 structures.
2. Filtered for only human α-β TCRs—264 structures.
3. Removed any structures with a resolution higher than 3 Å—220 structures.
4. Applied a redundancy filter for 95% CDR3β sequence identity—41 structures.

4.3 | Amino acid frequency calculation

For each position within the CDR1, CDR2, and CDR2.5 regions, the frequency of each amino acid residue was calculated separately for CD4+ and CD8+ TCRs. This was accomplished by aligning all collected sequences according to the IMGT numbering and counting the occurrence of each of the 20 standard amino acids at every position. The counts were then normalized by the total number of sequences in each category (MHC-I or MHC-II) to obtain relative frequencies.

Shannon entropy was computed for each position to quantify variability in amino acid distribution

separately for MHC class I and class II datasets. Amino acid frequencies at each position were calculated, excluding residues with zero frequency, and entropy values were derived accordingly.

4.4 | Statistical evaluation

Statistical tests were conducted to determine the significance of the differences in amino acid frequencies between the datasets. Initially, the normality of the distributions was assessed using the Shapiro–Wilk test, followed by Levene's test for homogeneity of variances. Depending on the outcome of these tests, either a standard independent *t*-test, Welch's *t*-test (for unequal variances), or a non-parametric Mann–Whitney *U* test was applied. The Bonferroni correction was used to adjust the *p*-values to account for multiple comparisons. The correction factor N for the Bonferroni Correction was set based on the number of different amino acids present at the same position since this corresponds to the number of tests conducted. Significant differences, after correction, were indicated on the plots with an asterisk above the corresponding bars.

4.5 | Making the predictor

Initially, training sequences were split into two groups based on their output labels (CD4+ or CD8+). The frequency of each amino acid at every position in the sequence was computed separately for each group. This resulted in two matrices: one for each group, showing the position-wise frequency of each amino acid.

Following this, a frequency ratio matrix was created by calculating the log of the division from the frequency of each amino acid at every position for the first group by its corresponding frequency in the second group. The natural logarithm helps scale down the values. These ratios were then used to replace the actual amino acid letters in the training and testing sequences, transforming the sequences into a matrix of frequency ratios.

4.6 | Weighted system

To systematically identify TCR residues in direct contact with MHC molecules, we first queried the PDB for all available crystal structures of human α-β TCR–pMHC complexes, including MHC class I and II. We retained only those structures with a resolution of ≤3.0 Å, thereby ensuring sufficiently high structural quality. Next, we removed duplicates and near-duplicates by applying a <95% sequence identity threshold to the TCR chains. After the filters, we have a non-redundant set of 41 unique complexes (29 involving MHC class I and 12 involving MHC class II).

For each TCR–pMHC structure, the CDR 1 and CDR2 loops were assessed for contacts with the MHC heavy chain. A TCR residue was defined as in contact if any heavy atoms lay within 4.0 Å of an MHC heavy chain atom. In addition, hydrogen bonds were recorded when donor-acceptor distances and angles satisfied standard criteria for hydrogen bonding, and salt bridges were defined by the presence of oppositely charged side chains within 4.0 Å. These contact definitions allowed for a detailed mapping of interface residues and their potential roles in TCR recognition. Positions involved in interactions of the analyzed crystals were retrieved. To underscore their importance, we amplified their impact through a weighted score. This score was calculated by multiplying the natural logarithm frequency ratio by a value ranging from 1 to 5, directly derived from the frequency of contacts in TCR:pMHC structures on these positions.

4.7 | Cross-validation setup

We used the repeated stratified five-fold cross-validation method to split our data into training and testing. This approach involves dividing the dataset into five distinct folds, ensuring that each fold represents the overall class distribution of the data (i.e., stratification). Specifically, the dataset is split so that each fold contains approximately the same percentage of samples of each target class as the complete set. The cross-validation process is then repeated five times, resulting in a total of 25 evaluations. In each iteration, the model is trained on four folds and validated on the remaining fold, rotating through all possible combinations. This repeated stratification improves the accuracy of our predictors.

4.8 | Machine learning classifiers

Three machine learning classifiers were selected for the study; we used the GridSearch function to find the best hyperparameters:

Linear support vector machine: ($C = 0.1$, class_weight = "balanced," dual = false, loss = "squared_hinge," max_iter = 800,000, penalty = "l2," tol = 1e−06).

Logistic regression: (class_weight = None, max_iter = 160,000, penalty = "none," solver = "lbfgs").

Random forest classifier: (bootstrap = False, class_weight = None, max_depth = 10, max_features = "auto," min_samples_leaf = 4, min_samples_split = 5, n_estimators = 800).

4.9 | Positions importance and coefficient analysis in predictive models

We studied the coefficients assigned to each position from the linear SVC and logistic regression models. These coefficients represent the weights used to define the linear decision boundary. Positive coefficients indicated that increasing the value of a feature made the model more likely to predict the MHC-I class, while negative coefficients had the opposite effect. The size of the number of each coefficient reflected the position's influence in the model, with larger absolute values indicating a stronger effect on the decision boundary.

On the other hand, the random forest classifier evaluated position importance by calculating how much each feature reduced impurity (e.g., Gini impurity or entropy) across the decision trees. Positions with higher values played a more significant role in making predictions, while those with lower values were less influential. Unlike coefficients, these importance values were non-negative and provided a relative measure of each position's contribution to the model's performance.

4.10 | Model evaluation

Model performance was evaluated using the ROC curve, which plots the TPR against the FPR. The AUC was also calculated for a scalar estimate of each classifier's performance. Averaged AUC values from all cross-validation folds were used as a consolidated performance metric. We also computed the F1-score, along with precision and recall metrics.

To complement our model evaluation strategy, we compare different classification approaches. The first script applies a "leave-one-(α + β)-gene-out" (LO(AB)O) cross-validation scheme, ensuring that samples sharing the same α- and β-gene combinations are excluded together in each fold to mitigate data leakage and overfitting. The second script compares a CDR-based feature representation against a V-gene-only approach through repeated stratified *k*-fold cross-validation.

4.11 | Sequence comparison

We extracted the CDR1 and CDR2 from the TCR sequences, concatenating the amino acids from the α and β chains while excluding gaps. We constructed a BLAST database using the CD8+ training sequences and queried the test sequences against it using BLASTp, retrieving only the top-scoring hit for each query. The *e*-values from the BLAST results were transformed into similarity scores by taking the negative logarithm, making the higher scores indicate higher sequence similarity. We then evaluated the predictive performance of this approach by generating ROC curves and calculating the AUC.

4.12 | Code availability

The code used to develop and evaluate the TCR/MHC prediction models and the associated training and testing datasets is available for public access at https://github.com/Marcus-Mendes/tcr-mhc_predictor. In the future, this tool will also be implemented on the IEDB website, making it accessible to broader users.

AUTHOR CONTRIBUTIONS

Marcus De Almeida Mendes: Conceptualization; investigation; writing – original draft; methodology; validation; visualization; writing – review and editing; data curation; formal analysis; software. **Leila Chihab:** Validation; methodology. **Jonas Birkelund Nilsson:** Writing – review and editing; validation; data curation. **Lonneke Scheffer:** Validation; writing – review and editing; data curation; methodology; software. **Morten Nielsen:** Writing – review and editing; supervision. **Bjoern Peters:** Funding acquisition; investigation; conceptualization; writing – review and editing; supervision; project administration; validation; resources.

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DATA AVAILABILITY STATEMENT

The data supporting this study's findings are available in Immune Epitope Database (www.iedb.org) and VDJdb (vdjdb.cdr3.net).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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